Quadruples sharing their largest element

Problèmes ouverts — A. Dujella · 2026

Problem 4.3 (Dujella, 2026). Are there infinitely many pairs of Diophantine quadruples with the same largest element ?

Remark. Gibbs (1999) found three such pairs : $\{1, 16640, 16899, 1124864520\}$ and $\{1, 399, 703920, 1124864520\}$, $\{120, 145673, 154155, 10778986831096\}$ and $\{8, 120, 2805566778, 10778986831096\}$, $\{528, 3253511, 3336933, 229294$ and $\{15, 528, 723737525421, 22929452257545400\}$. Dujella (2026) found the fourth example : $\{528, 1470, 227476966$ and $\{528, 1829619951, 1827654733, 7062345445140145061720\}$.